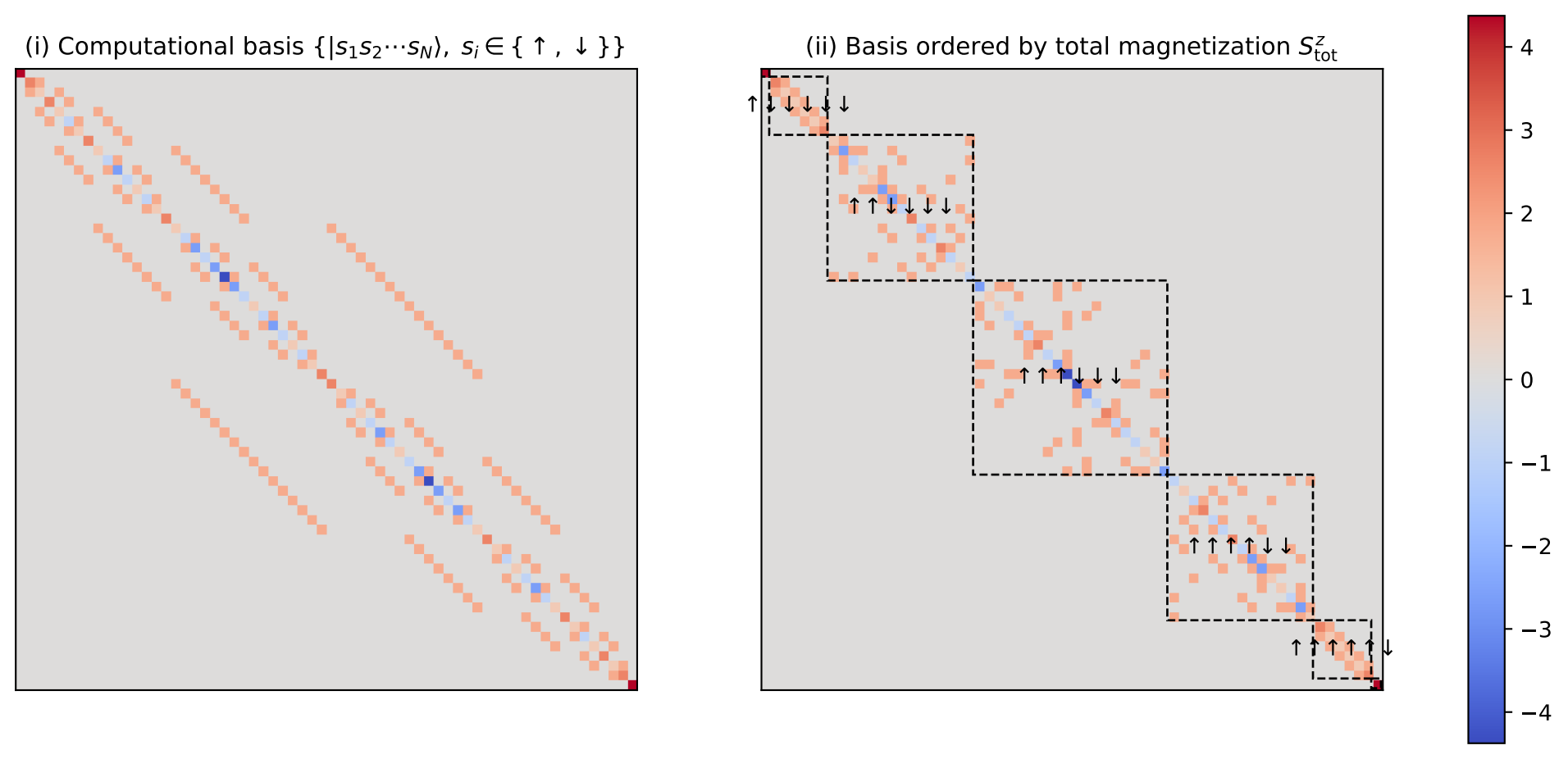
(i) Computational basis $\{|s_1 s_2 \cdots s_N\rangle,\ s_i \in \{\uparrow, \downarrow\}\}$
(ii) Basis ordered by total magnetization S^z_{tot}
↑↓↓↓↓↓
↑↑↓↓↓↓
↑↑↑↓↓↓
↑↑↑↑↓↓
↑↑↑↑↑↓
4
3
2
1
0
−1
−2
−3
−4